

Today, in this class..



Metabolism



ATP energy

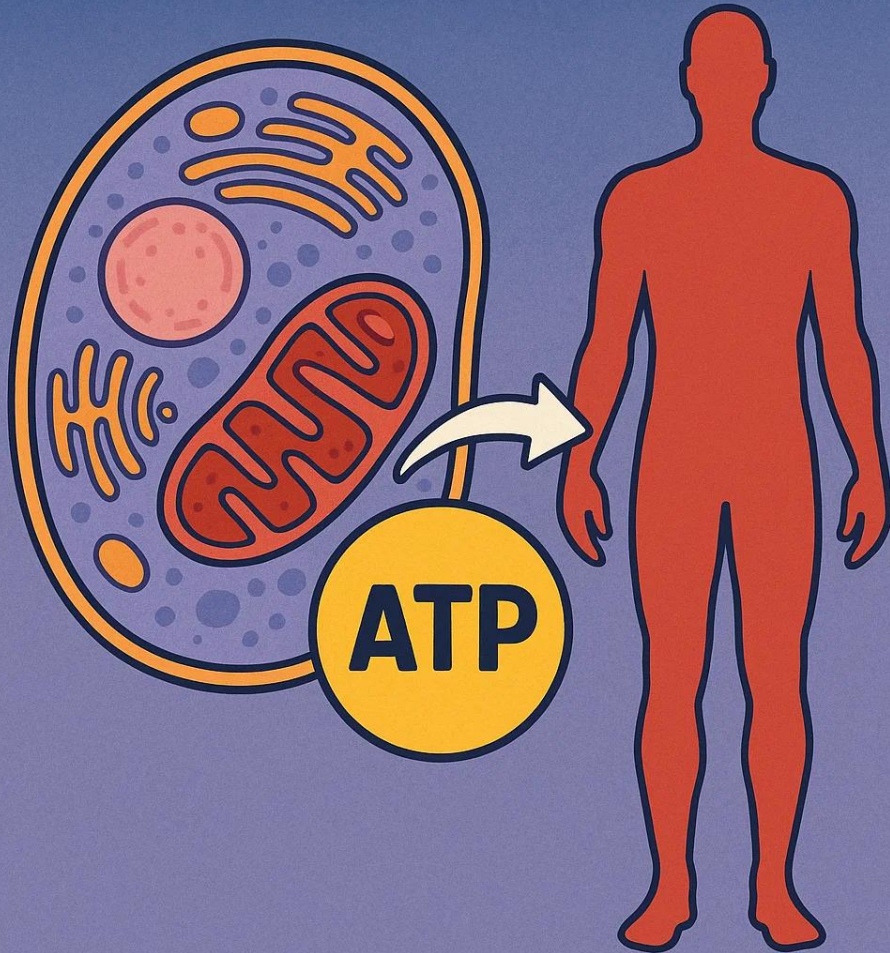


Reaction
spontaneity

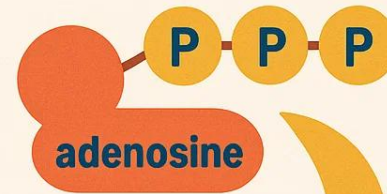


How ATP works

ATP ENERGY IN CELLS



ATP

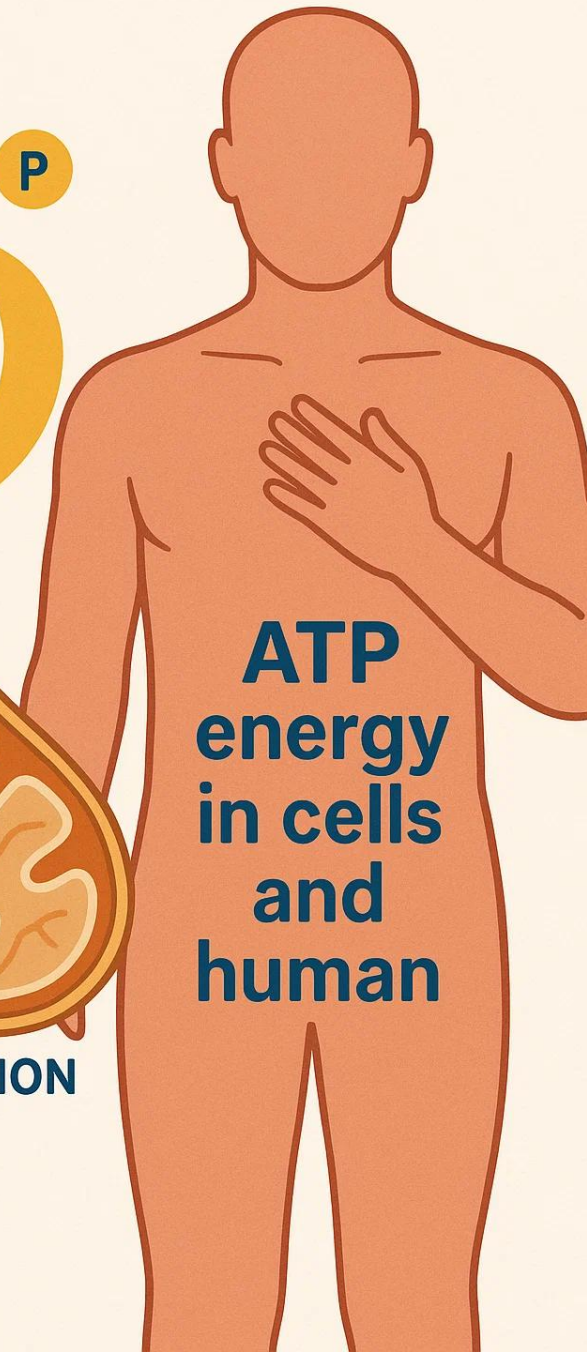


ENERGY



MITOCHONDRION

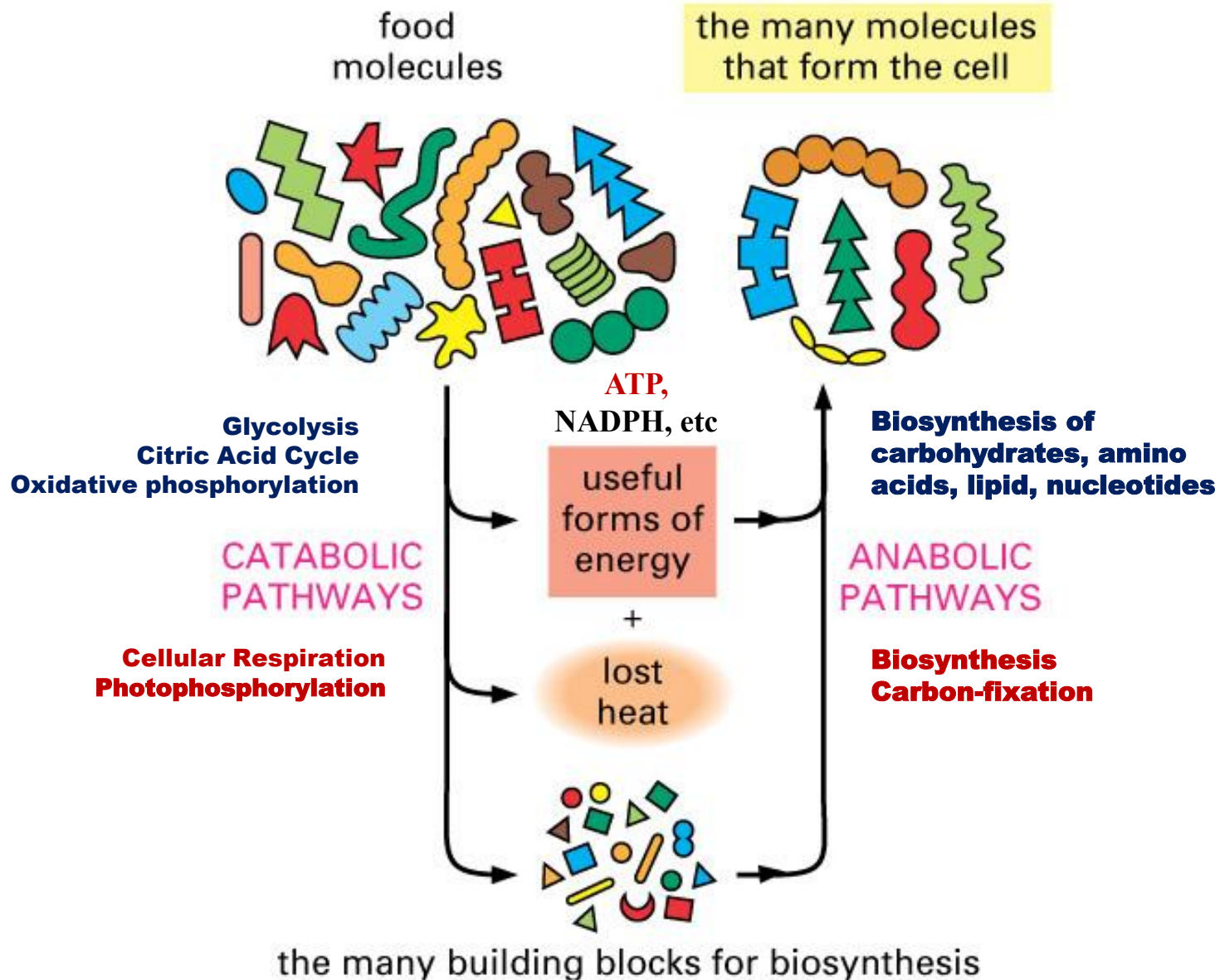
ATP
energy
in cells
and
human



PART II Ch13 ~ Ch23

Biological Energy & Metabolism

Overview of the PART II (chapter 13 ~ chapter 23)

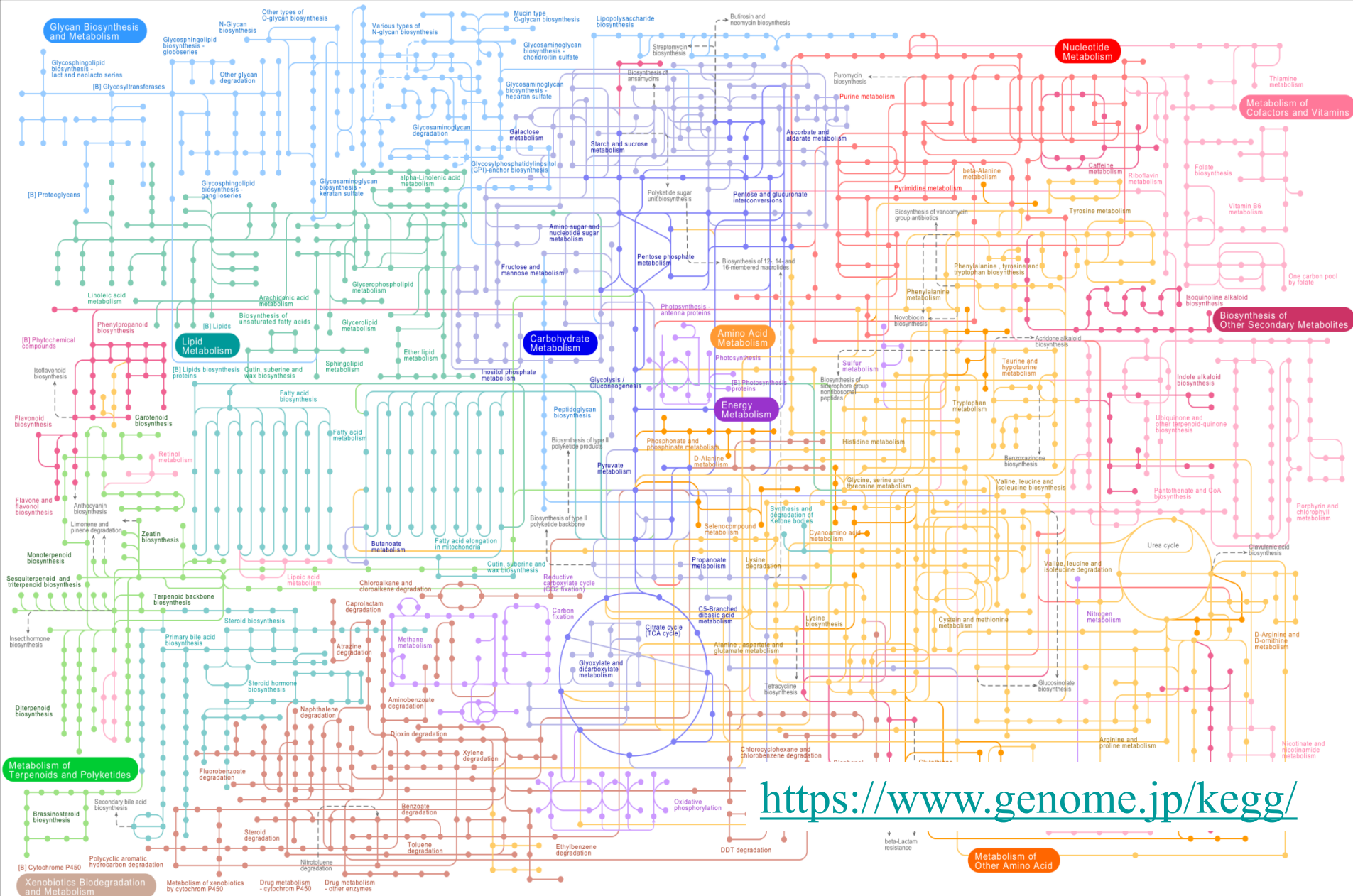


What is Metabolism ?

Metabolism is **the sum** of all the **chemical transformations** taking place in a cell or organism.

Metabolism occurs through **a series of enzyme-catalyzed reactions** that constitute metabolic pathways.

Complexity of Metabolism



<https://www.genome.jp/kegg/>

The Purpose of Metabolism ?

Metabolism is highly coordinated cellular activities,

- (1) To obtain **chemical energy**
- (2) To **convert nutrient molecules** into the cell's own molecules
- (3) To polymerize monomeric precursors **into macromolecules**

 **Energy Generation & Biosynthesis**

David L. Nelson and Michael M. Cox

LEHNINGER

PRINCIPLES OF BIOCHEMISTRY

Sixth Edition

CHAPTER 13

**Bioenergetics and Biochemical
Reaction Types**

Biochemical Energy

ATP

= Energy currency

= Stored energy

How **ATP can store and provide **Energy** ?**



Energy states and flows



Stored E

Used E

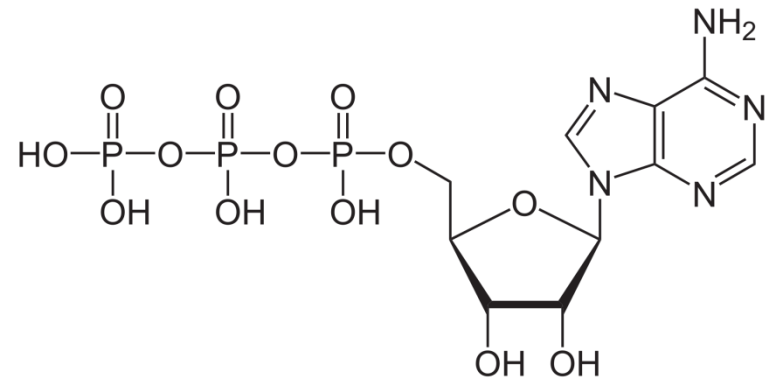
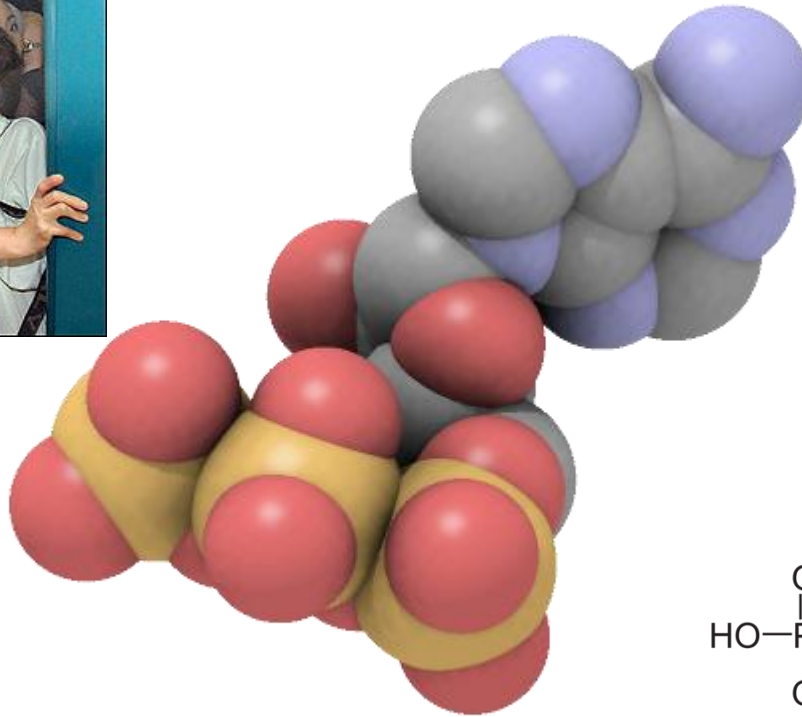
unstable

more stable

high E state

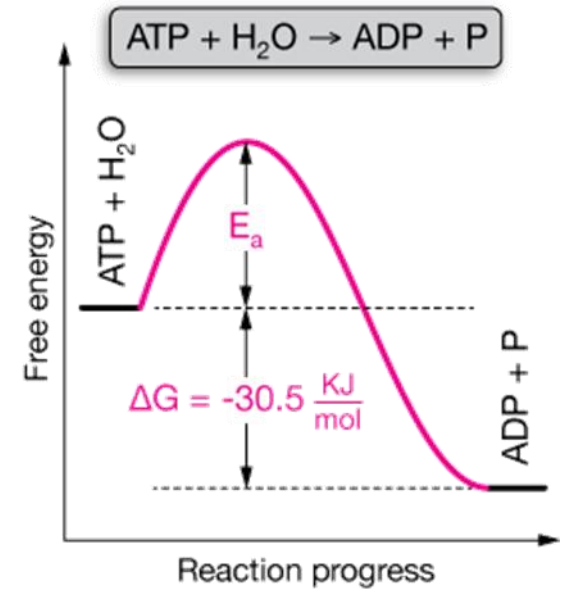
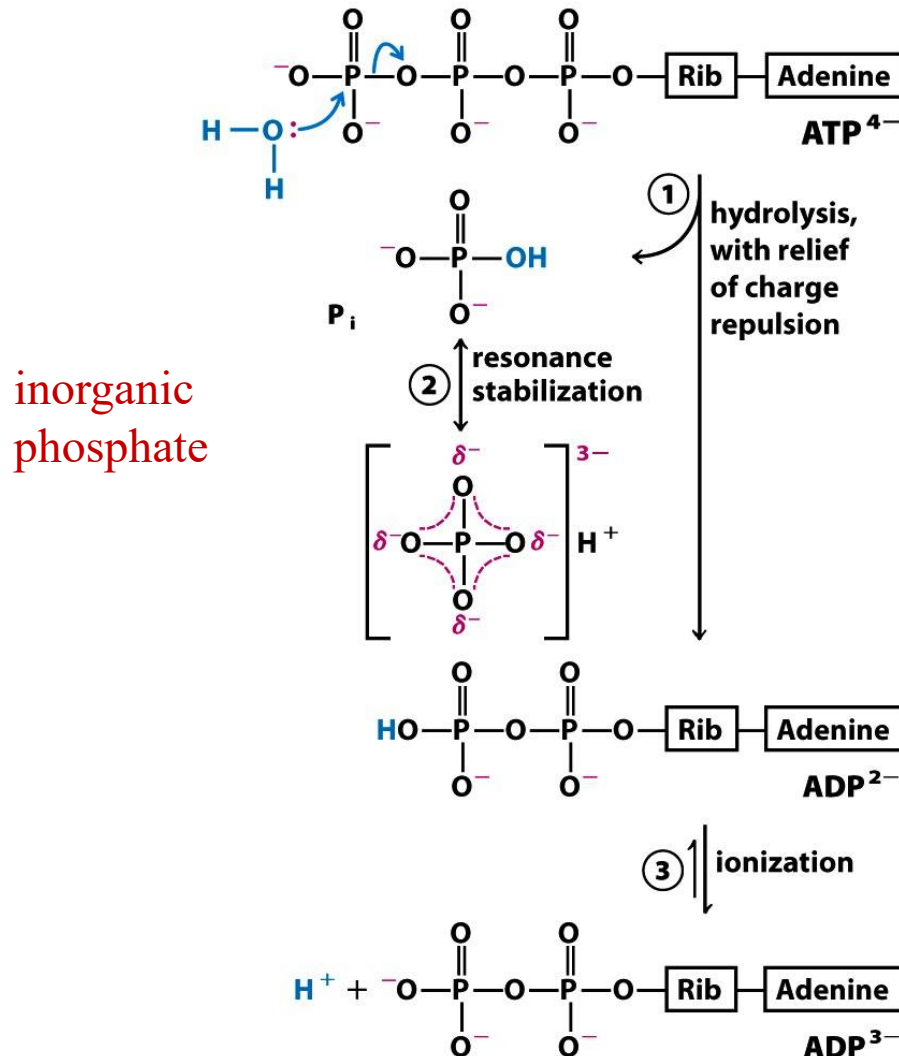
low E state

ATP has high potential Energy



It is too crowded !

Large free-energy change associated with ATP hydrolysis



④ solvation of P_i and ADP



Figure 13-11
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Moreover, because the product concentration in the cell is very low, mass action favors the hydrolysis in the cell

TABLE 13–5	Adenine Nucleotide, Inorganic Phosphate, and Phosphocreatine Concentrations in Some Cells				
	Concentration (mM)*				
	ATP	ADP [†]	AMP	P _i	PCr
Rat hepatocyte	3.38	1.32	0.29	4.8	0
Rat myocyte	8.05	0.93	0.04	8.05	28
Rat neuron	2.59	0.73	0.06	2.72	4.7
Human erythrocyte	2.25	0.25	0.02	1.65	0
<i>E. coli</i> cell	7.90	1.04	0.82	7.9	0

- The actual free energy of hydrolysis of ATP, phosphorylation potential ΔG_p , can be calculated

$$\Delta G_p = \Delta G'^{\circ} + RT \ln \frac{[\text{ADP}] [\text{P}_i]}{[\text{ATP}]} = -30.5 - 22 \text{ kJ/mol}$$

(in erythrocyte)

Gibbs free energy, G

$$\Delta G = \Delta H - T\Delta S$$

G: Gibbs free energy – the amount of energy capable of doing work. **If ΔG is negative, the energy is released**

H: Enthalpy – the heat constant of the reacting system

S: Entropy – randomness or disorderness

ΔG : free energy change

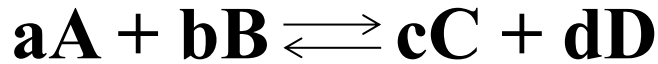
ΔG° : standard free energy change
at 298K, 1atm (101.3 kPa)

$\Delta G'^\circ$: Transformed standard free-energy change in biochemical reaction

Biochemical standard free-energy change

at 298K, 1atm, pH 7, $[H_2O]=55.5M$, $[Mg^{+2}]=1mM$

When initial conc. of each component is **1.0 M**,
it is a constant



$$K'_{eq} = \frac{[C]^c [D]^d}{[A]^a [B]^b} = \mathbf{Q} \text{ (mass-action ratio)}$$

$$\Delta G = \Delta G'^{\circ} + RT \ln \mathbf{Q} = \Delta G'^{\circ} + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

When equilibrium,

$$\Delta G = 0 = \Delta G'^{\circ} + RT \ln \frac{[C]_{eq} [D]_{eq}}{[A]_{eq} [B]_{eq}} = \Delta G'^{\circ} + RT \ln K'_{eq}$$

$$\Delta G'^{\circ} = - RT \ln K'_{eq}$$

$\Delta G'^{\circ}$ **biochemical standard free-energy change** is simply an alternative expression of equilibrium constant

ΔG : represents **how much it can work, and how far is the system from equilibrium**

TABLE 13–2**Relationship between Equilibrium Constants and Standard Free-Energy Changes of Chemical Reactions**

K'_{eq}	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)*
10^3	–17.1	–4.1
10^2	–11.4	–2.7
10^1	–5.7	–1.4
1	0.0	0.0
10^{-1}	5.7	1.4
10^{-2}	11.4	2.7
10^{-3}	17.1	4.1
10^{-4}	22.8	5.5
10^{-5}	28.5	6.8
10^{-6}	34.2	8.2

if we start with ...



Forward reaction



Reverse reaction

***Although joules and kilojoules are the standard units of energy and are used throughout this text, biochemists and nutritionists sometimes express $\Delta G'^{\circ}$ values in kilocalories per mole. We have therefore included values in both kilojoules and kilocalories in this table and in Tables 13–4 and 13–6. To convert kilojoules to kilocalories, divide the number of kilojoules by 4.184.**

Table 13-2

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Q

TABLE 13-4

Standard Free-Energy Changes of Some Chemical Reactions

Reaction type	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Hydrolysis reactions		
Acid anhydrides		
Acetic anhydride + H ₂ O $\longrightarrow$ 2 acetate	-91.1	-21.8
ATP + H ₂ O $\longrightarrow$ ADP + P _i	-30.5	-7.3
ATP + H ₂ O $\longrightarrow$ AMP + PP _i	-45.6	-10.9
PP _i + H ₂ O $\longrightarrow$ 2P _i	-19.2	-4.6
UDP-glucose + H ₂ O $\longrightarrow$ UMP + glucose 1-phosphate	-43.0	-10.3
Esters		
Ethyl acetate + H ₂ O $\longrightarrow$ ethanol + acetate	-19.6	-4.7
Glucose 6-phosphate + H ₂ O $\longrightarrow$ glucose + P _i	-13.8	-3.3
Amides and peptides		
Glutamine + H ₂ O $\longrightarrow$ glutamate + NH ₄ ⁺	-14.2	-3.4
Glycylglycine + H ₂ O $\longrightarrow$ 2 glycine	-9.2	-2.2
Glycosides		
Maltose + H ₂ O $\longrightarrow$ 2 glucose	-15.5	-3.7
Lactose + H ₂ O $\longrightarrow$ glucose + galactose	-15.9	-3.8
Rearrangements		
Glucose 1-phosphate $\longrightarrow$ glucose 6-phosphate	-7.3	-1.7
Fructose 6-phosphate $\longrightarrow$ glucose 6-phosphate	-1.7	-0.4
Elimination of water		
Malate $\longrightarrow$ fumarate + H ₂ O	3.1	0.8
Oxidations with molecular oxygen		
Glucose + 6O ₂ $\longrightarrow$ 6CO ₂ + 6H ₂ O	-2,840	-686
Palmitate + 23O ₂ $\longrightarrow$ 16CO ₂ + 16H ₂ O	-9,770	-2,338

Table 13-4

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Note

Q

1. The spontaneity of a reaction depends on ΔG , not $\Delta G'^{\circ}$
if we remove the product $\rightarrow -RT \ln Q > \Delta G'^{\circ}$
 $\rightarrow$ forward reaction
2. ΔG is a variable that depends on physical constant $\Delta G'^{\circ}$ and on the concentrations of reactants and products
3. ΔG is the maximum amount of free energy that the reaction theoretically deliver : some energy is lost to entropy
4. Activation energy is needed even in the thermodynamically favorable reactions : the role of enzymes in cell
5. Enzymes cannot change Eq. constants, but increase the rate of reaction
6. $\Delta G'^{\circ}$ values are additive
 K'_{eq} values are multiplicative

High energy state of ATP

Reaction spontaneity

The role of ATP

ATP energy can be used for group transfers

: ATP provides energy in two-step process

* ATP is kinetically stable, due to huge activation energies (200 to 400 kJ/mol)

In enzymes, the phosphate groups are stabilized by the surrounding positive charges
→ Reducing the activation energy

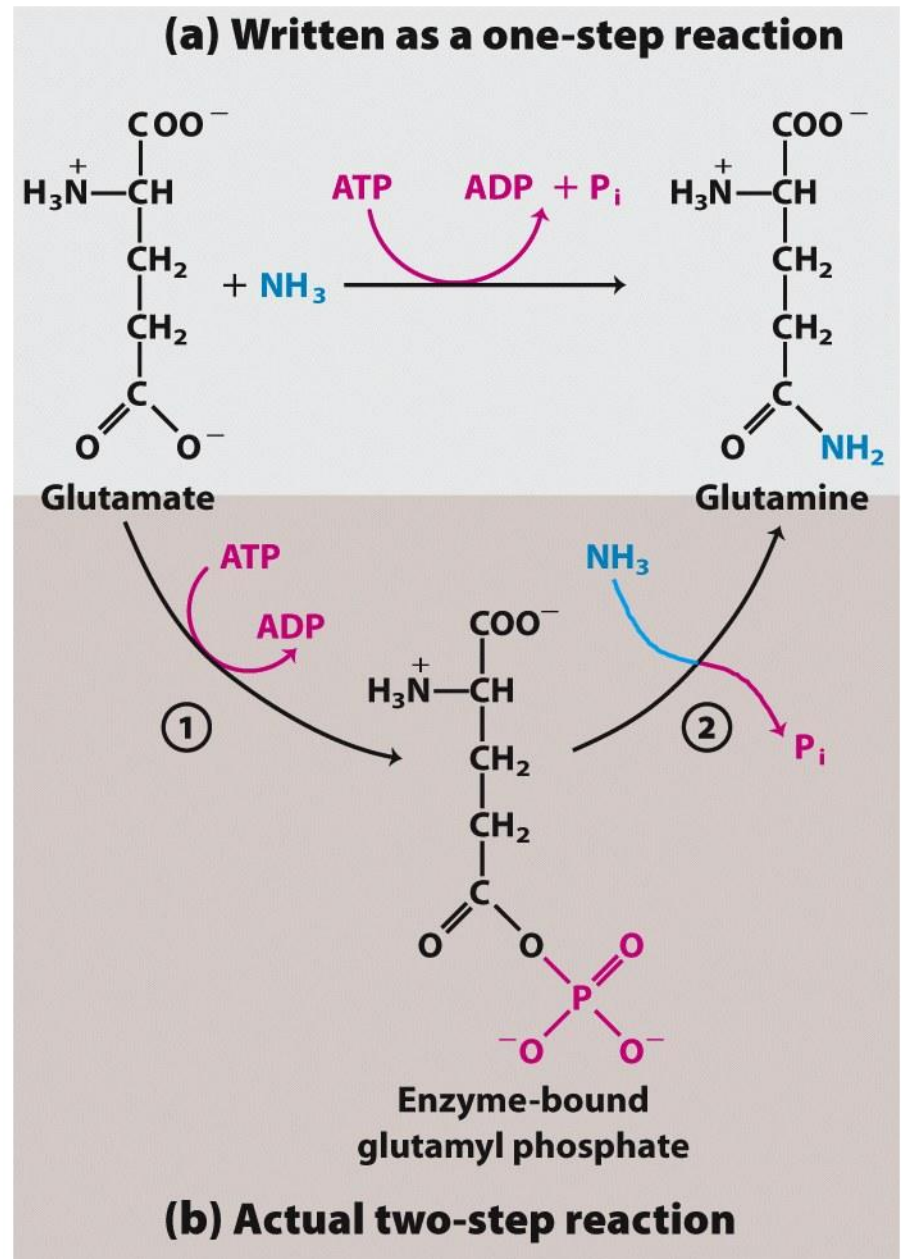


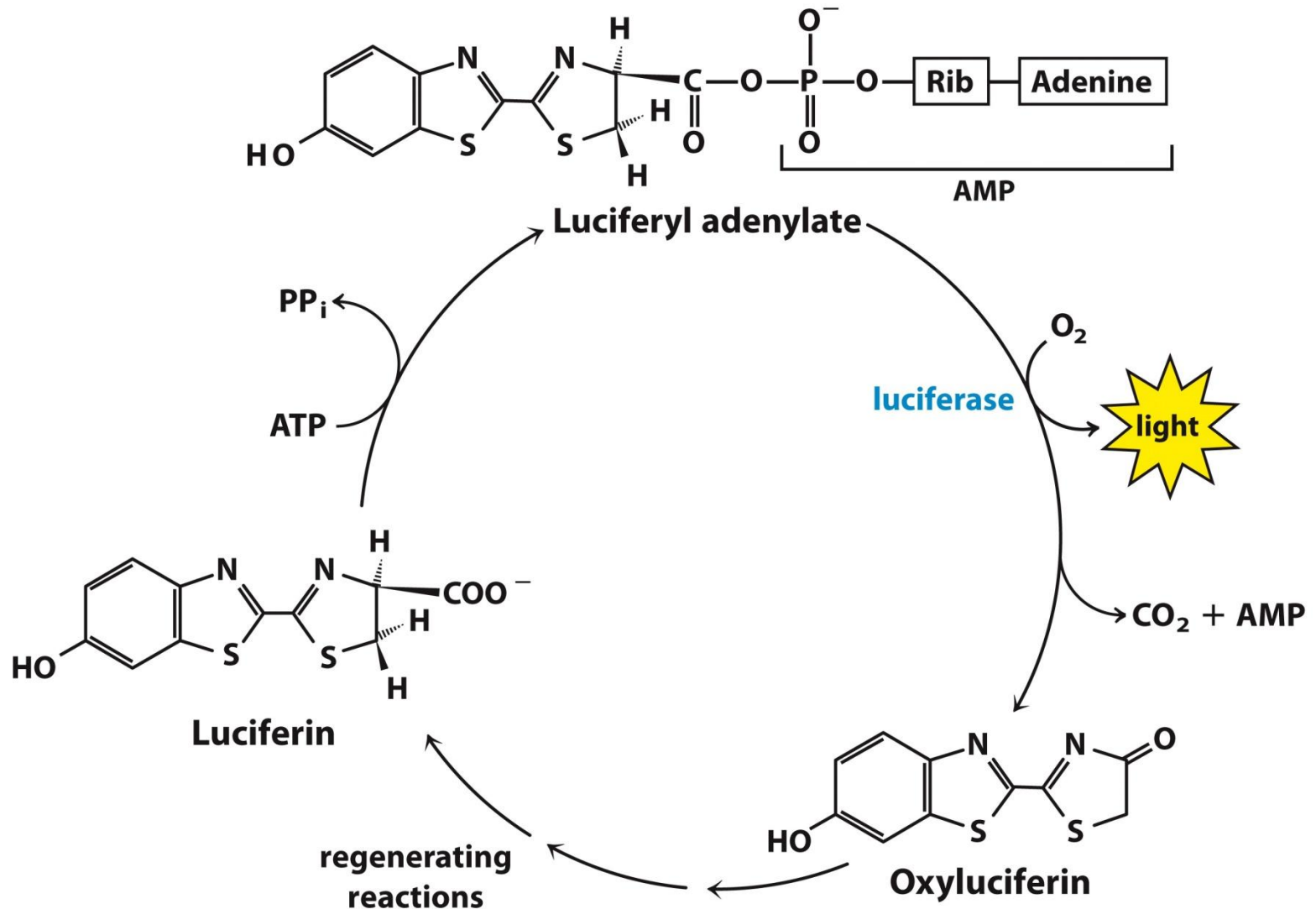
Figure 13-18

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ATP is needed for luciferase reaction:

Converting chemical energy to light energy



ATP energizes active transport by conformational change

Phosphorylation of Na⁺ K⁺ ATPase

→ Forces a change in protein conformation

Dephosphorylation

→ Return to the original conformation

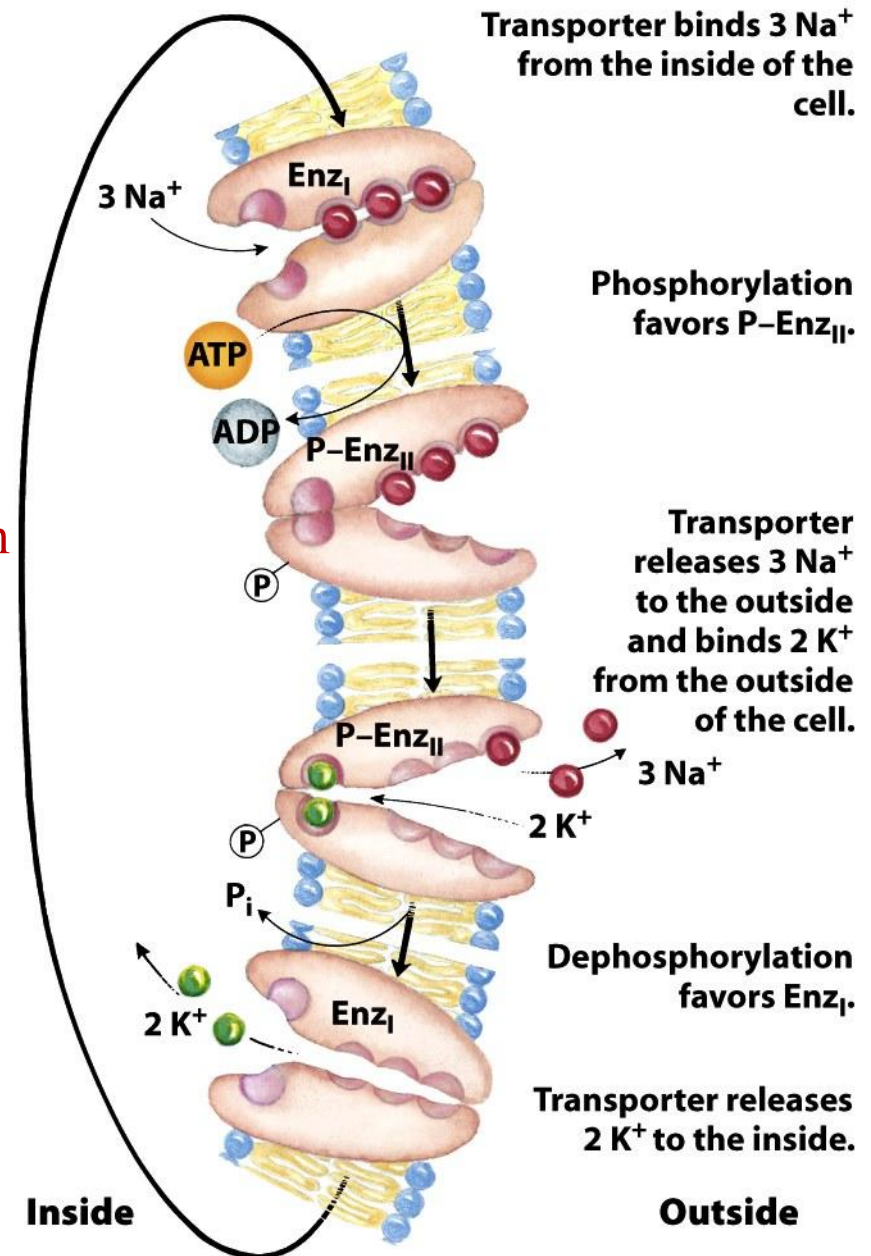


Figure 11-37

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ATP energizes muscle contraction

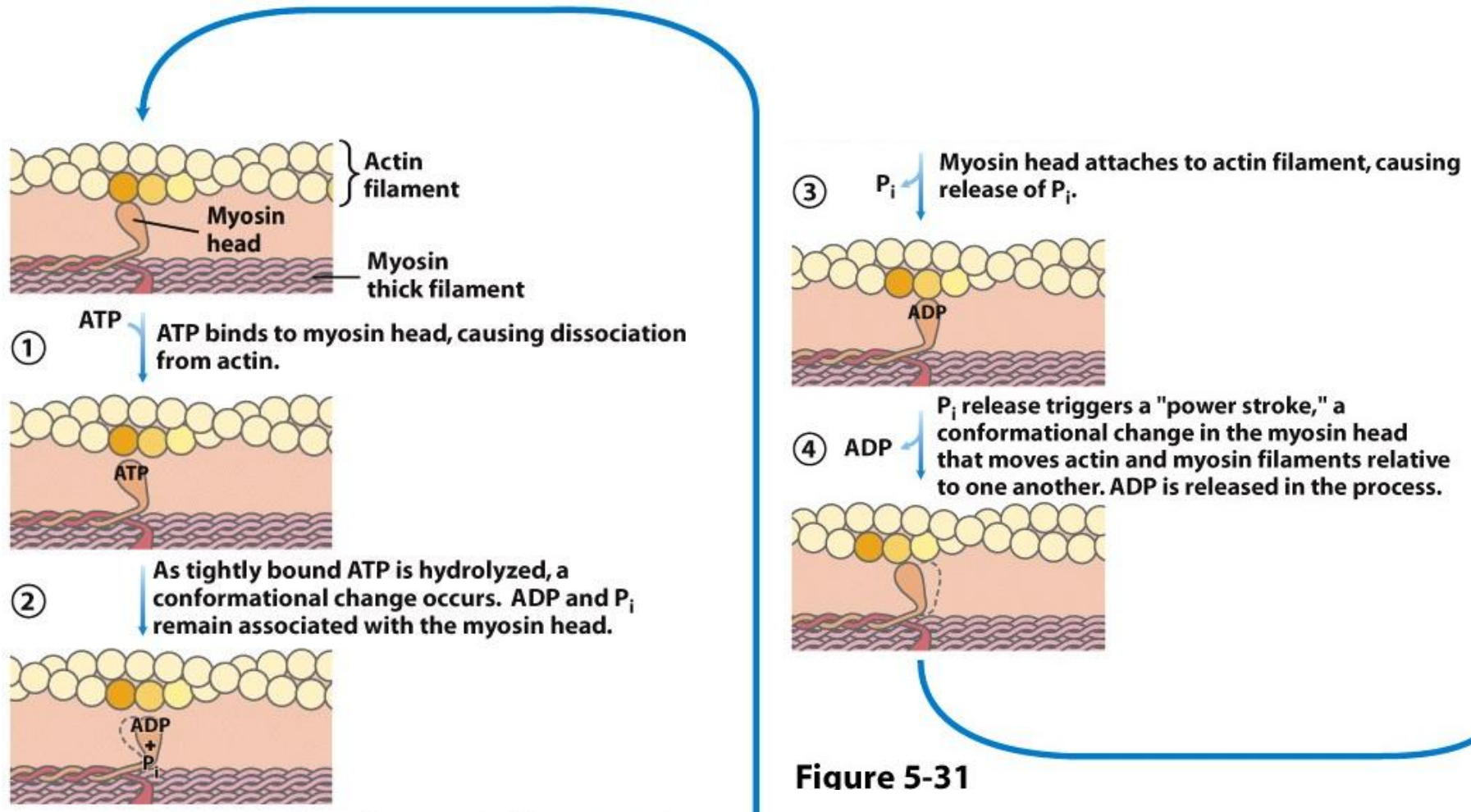


Figure 5-31

ATP hydrolysis **directly** provide chemical energy to change conformation

Some processes do involve direct hydrolysis of ATP

- Provide the energy to cycle between two conformations
- Producing mechanical motion

(examples)

Muscle contraction

Movement of enzymes along DNA

Movement of ribosomes along messenger RNA

Helicase, RecA, topoisomerase



Other energy sources ...

Hydrolysis of phosphoenolpyruvate (PEP)

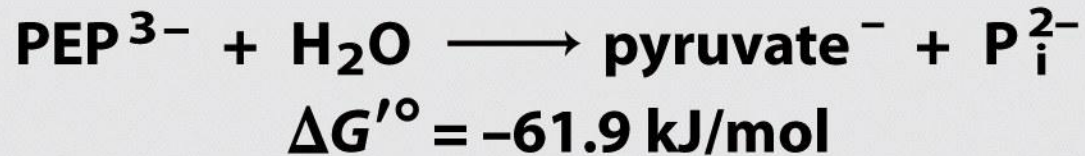
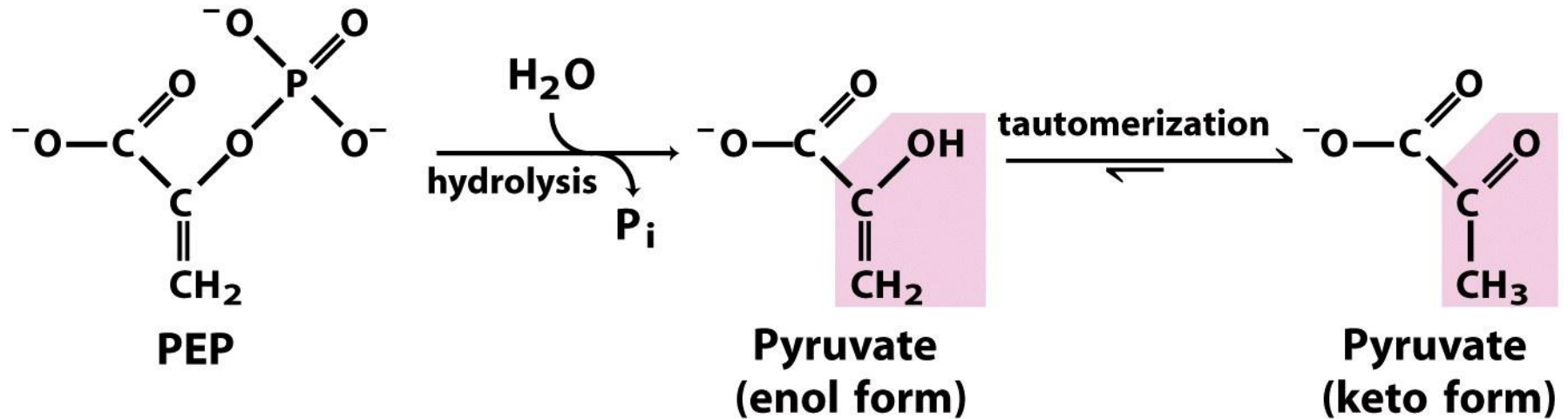


Figure 13-13

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Large free-energy change !

Hydrolysis of 1,3-bisphosphoglycerate

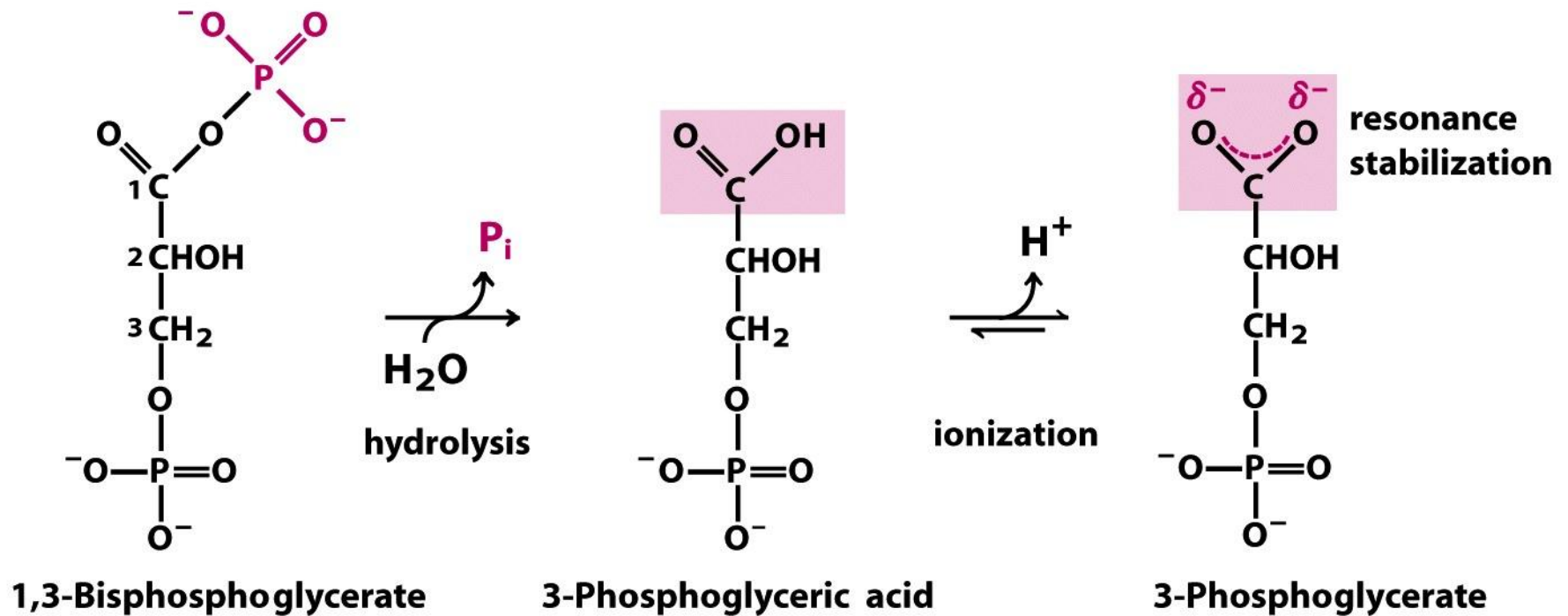


Figure 13-14

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Hydrolysis of phosphocreatine

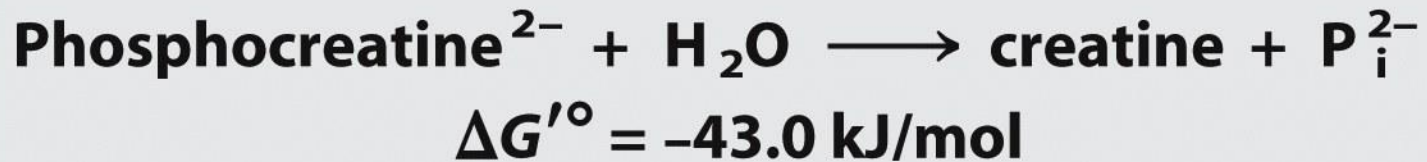
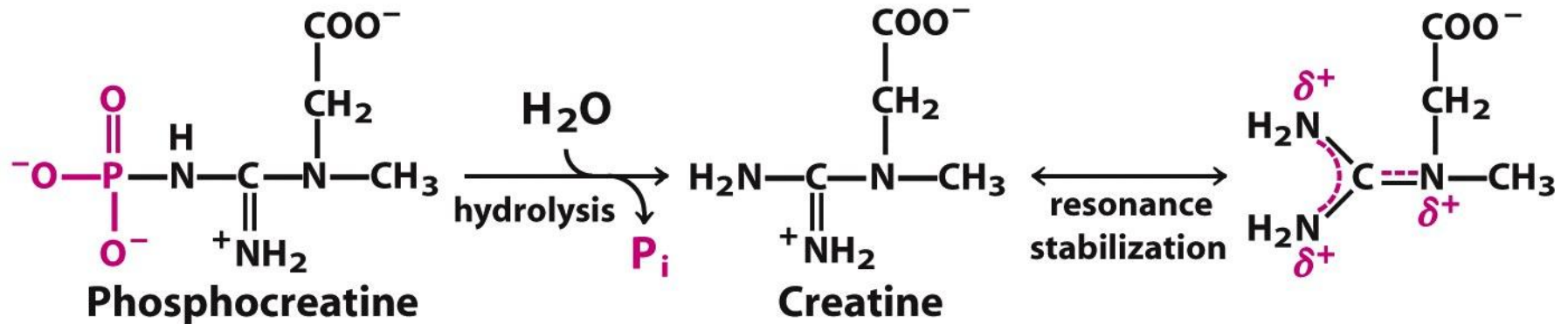
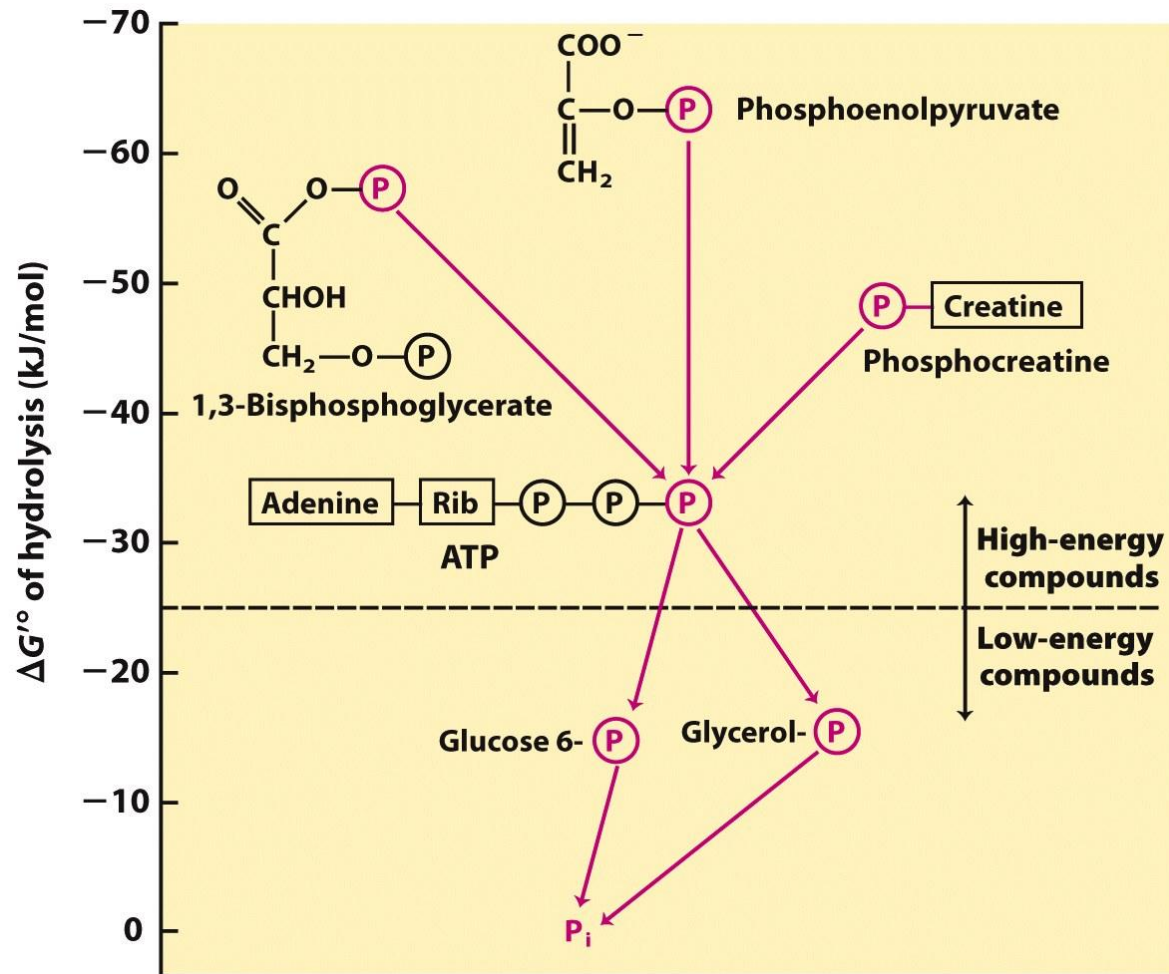


Figure 13-15

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Ranking of biological phosphate compounds by standard free energies of hydrolysis



Phosphorylated compound can be synthesized by coupling the synthesis to breakdown of another compound

Hydrolysis of thioesters

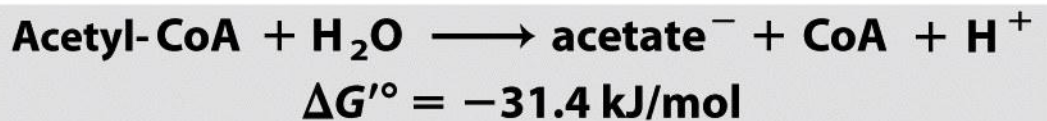
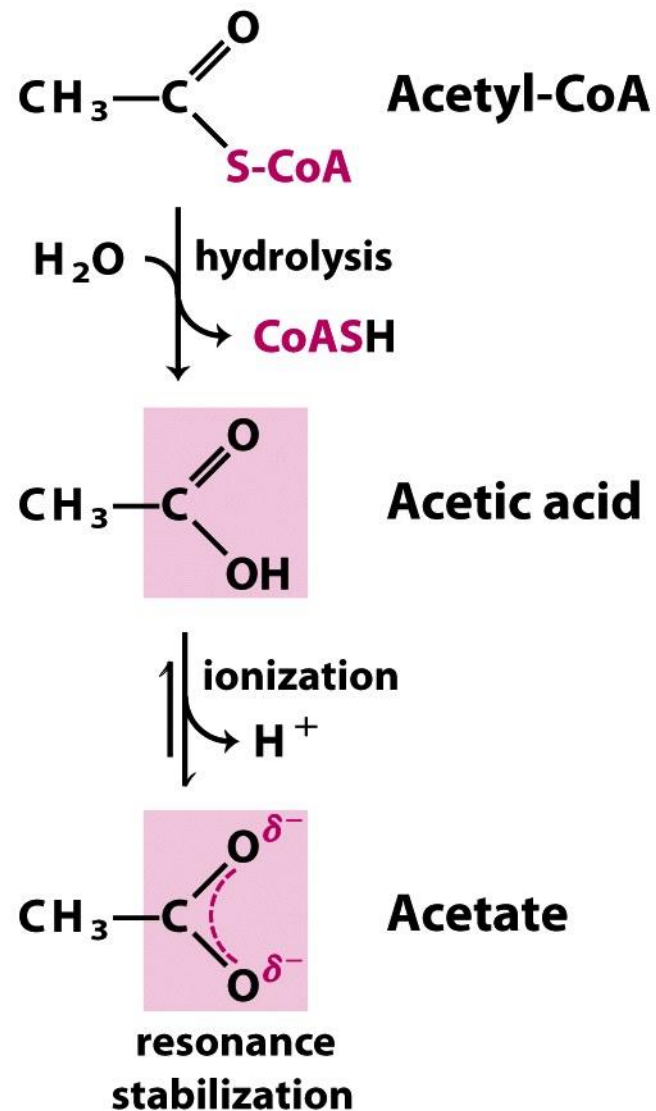


Figure 13-16

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Free energy of hydrolysis for thioesters and oxygen esters

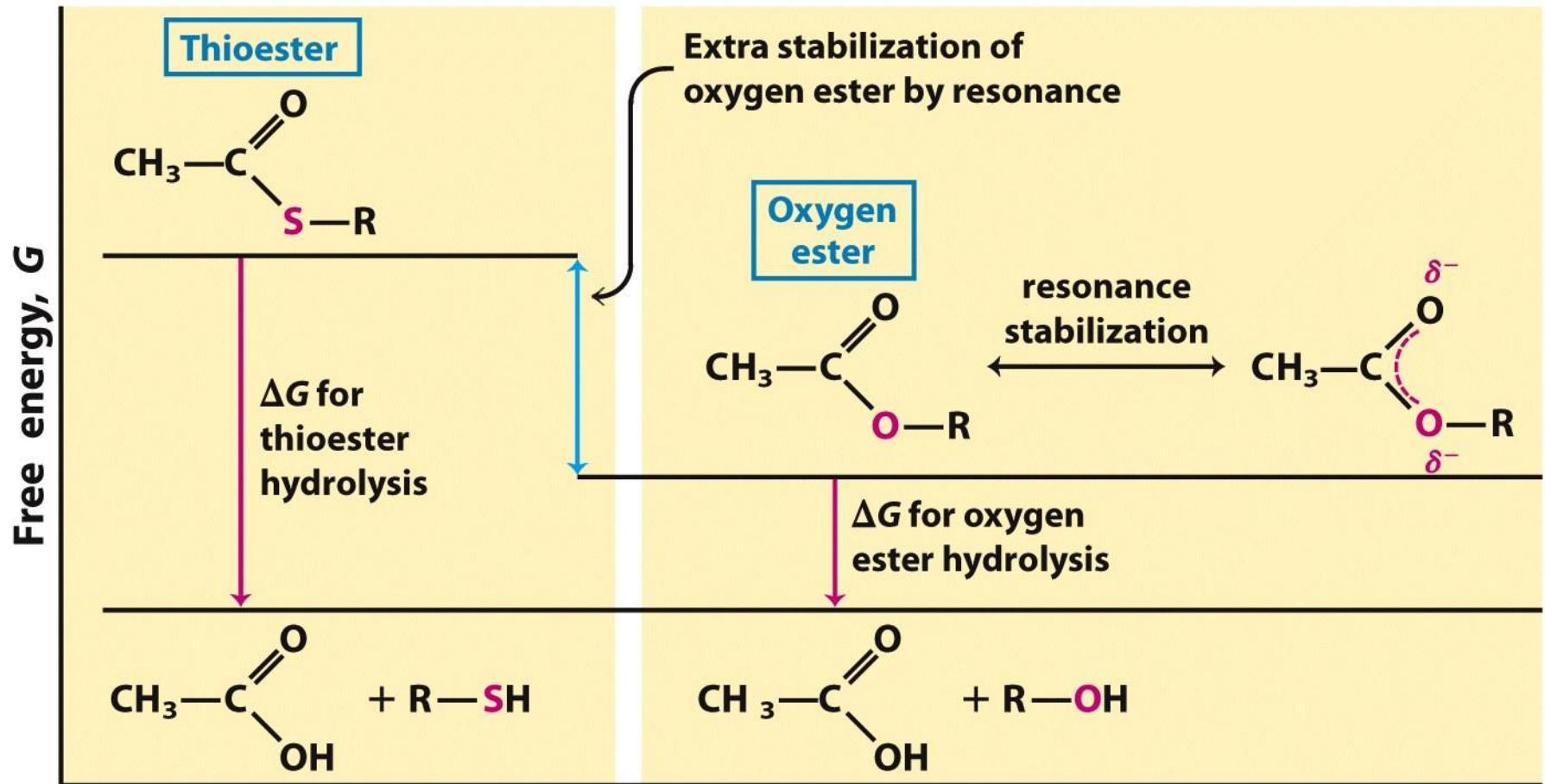


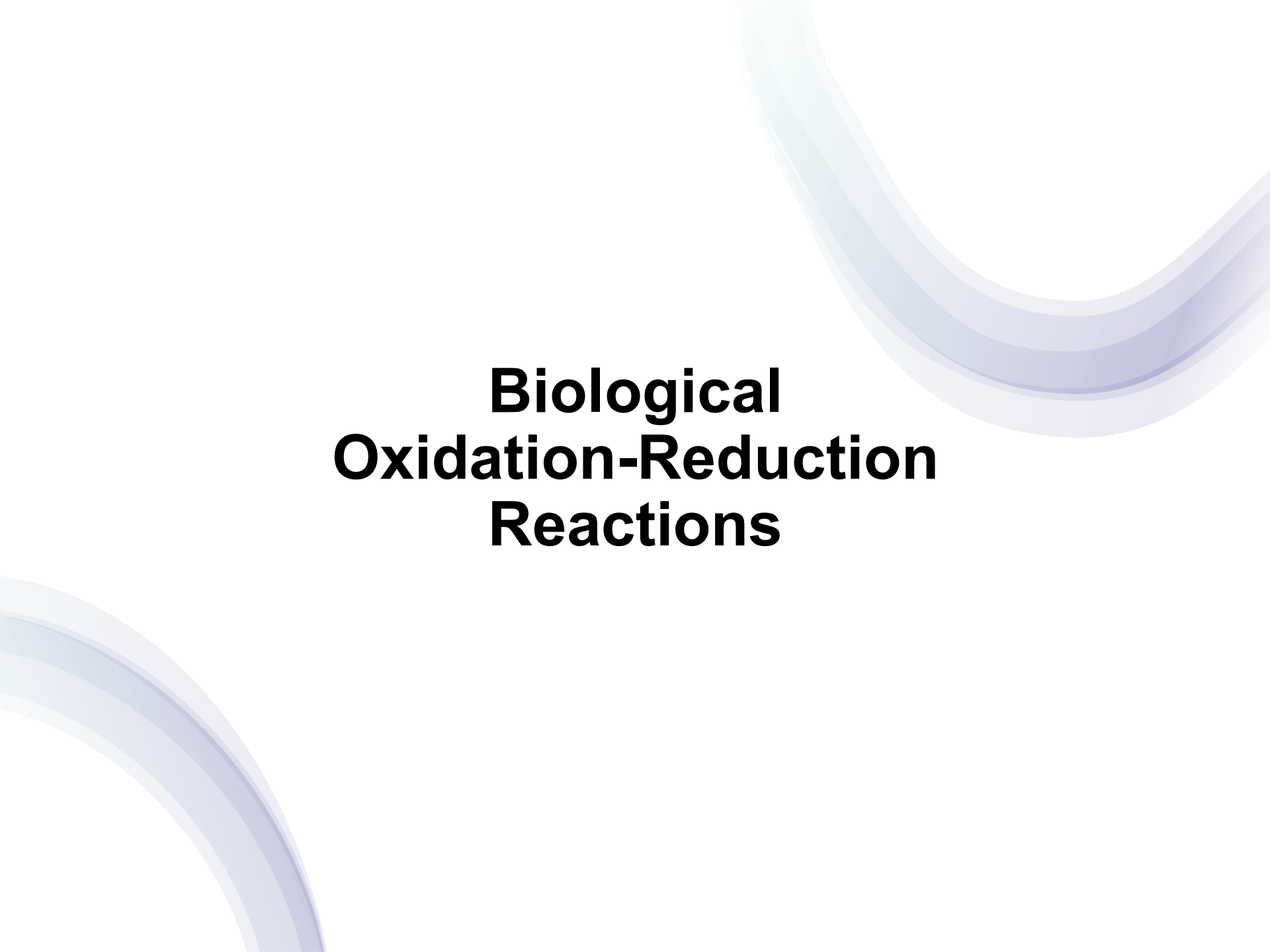
Figure 13-17

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Summary

- Chemical basis for the large free-energy change associated with ATP hydrolysis
- ATP provides energy by group transfers, not by simple hydrolysis
 - But, Some processes do involve direct hydrolysis of ATP
- Three positions on ATP for attack by the nucleophile
- Role of ATP energy (examples):
 - (1) Assembly of informational macromolecules
 - (2) ATP energizes active transport
 - (3) Role of ATP energy in muscle contraction



Biological Oxidation-Reduction Reactions

Oxidation

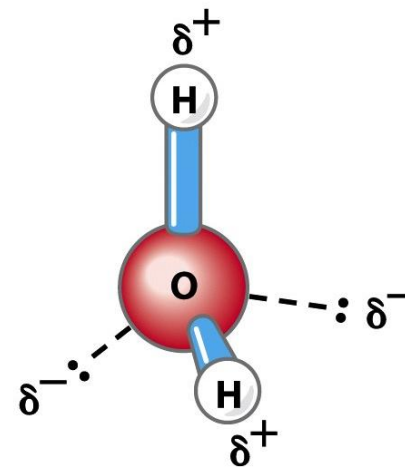
losing e-
losing H



H lost e-
H has been oxidized

Reduction

gaining e-
gaining H



O gained e-
O has been reduced

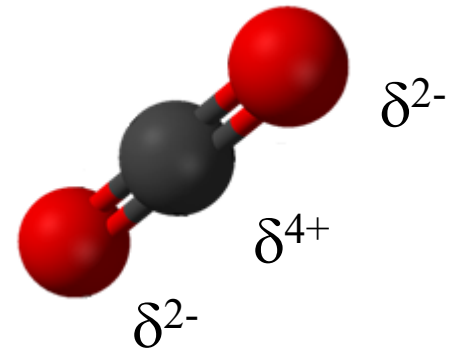
Biological Oxidation & Reduction



C-H

O=C=O

H-O-H



C lost H

C lost e-

C has been oxidized

O gained H

O gained e-

O has been reduced



Oxidation states of carbon in the biosphere

Q

Methane		8	Acetone (ketone)		2
Ethane (alkane)		7	Formic acid (carboxylic acid)		2
Ethene (alkene)		6	Carbon monoxide		2
Ethanol (alcohol)		5	Acetic acid (carboxylic acid)		1
Acetylene (alkyne)		5	Carbon dioxide		0
Formaldehyde		4	Numbers: no of electrons own by red carbon		
Acetaldehyde (aldehyde)		3			

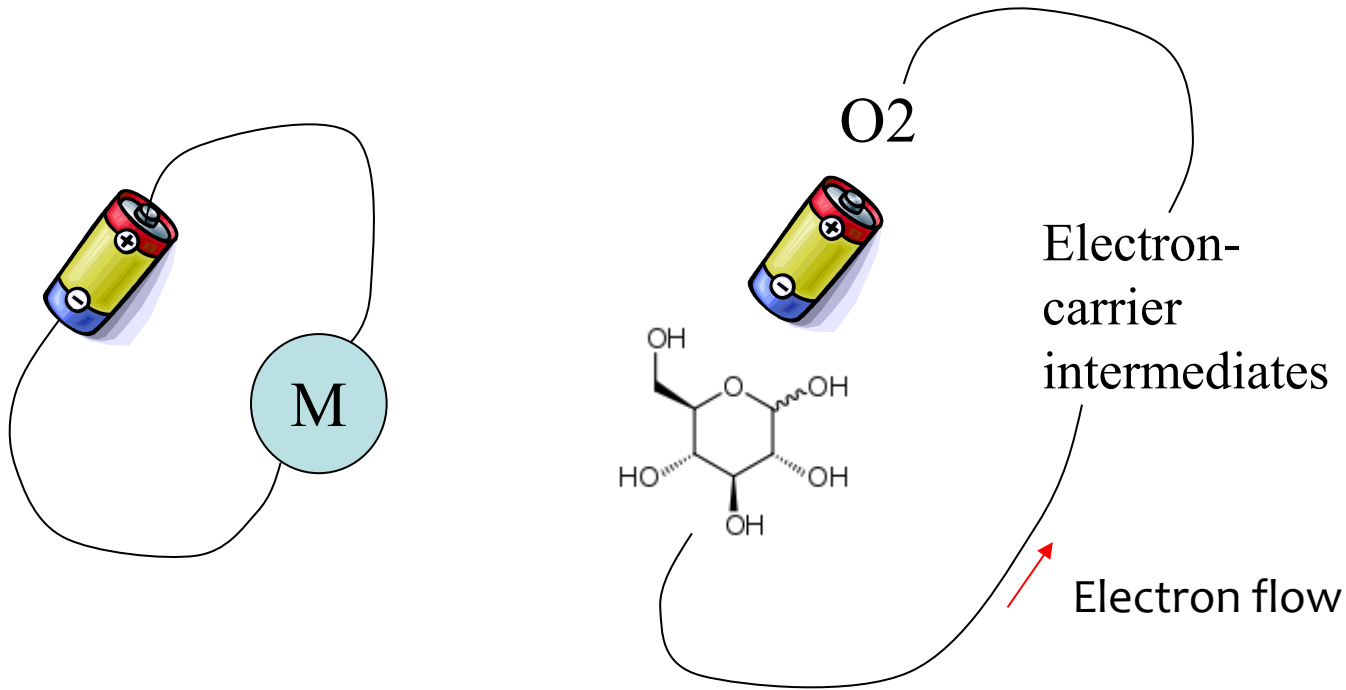
Numbers: no of electrons own by red carbon

Electronegativity:
H < C < S < N < O

Figure 13-22
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How the movement of electron can provide energy ?

Flow of electrons can do biological work

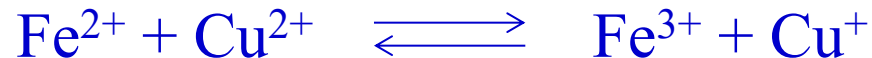


Difference in electron affinity : electromotive force, emf

→ Provide energy (ex. transmembrane pH difference, transmembrane electrical potential, ATP synthesis, flagellar motion, etc)

Four ways of electron transfer

(1) Directly as electrons



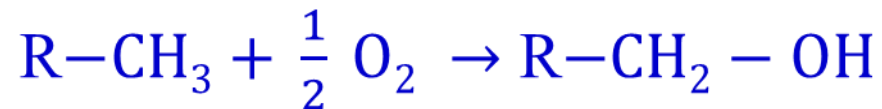
(2) As hydrogen atoms



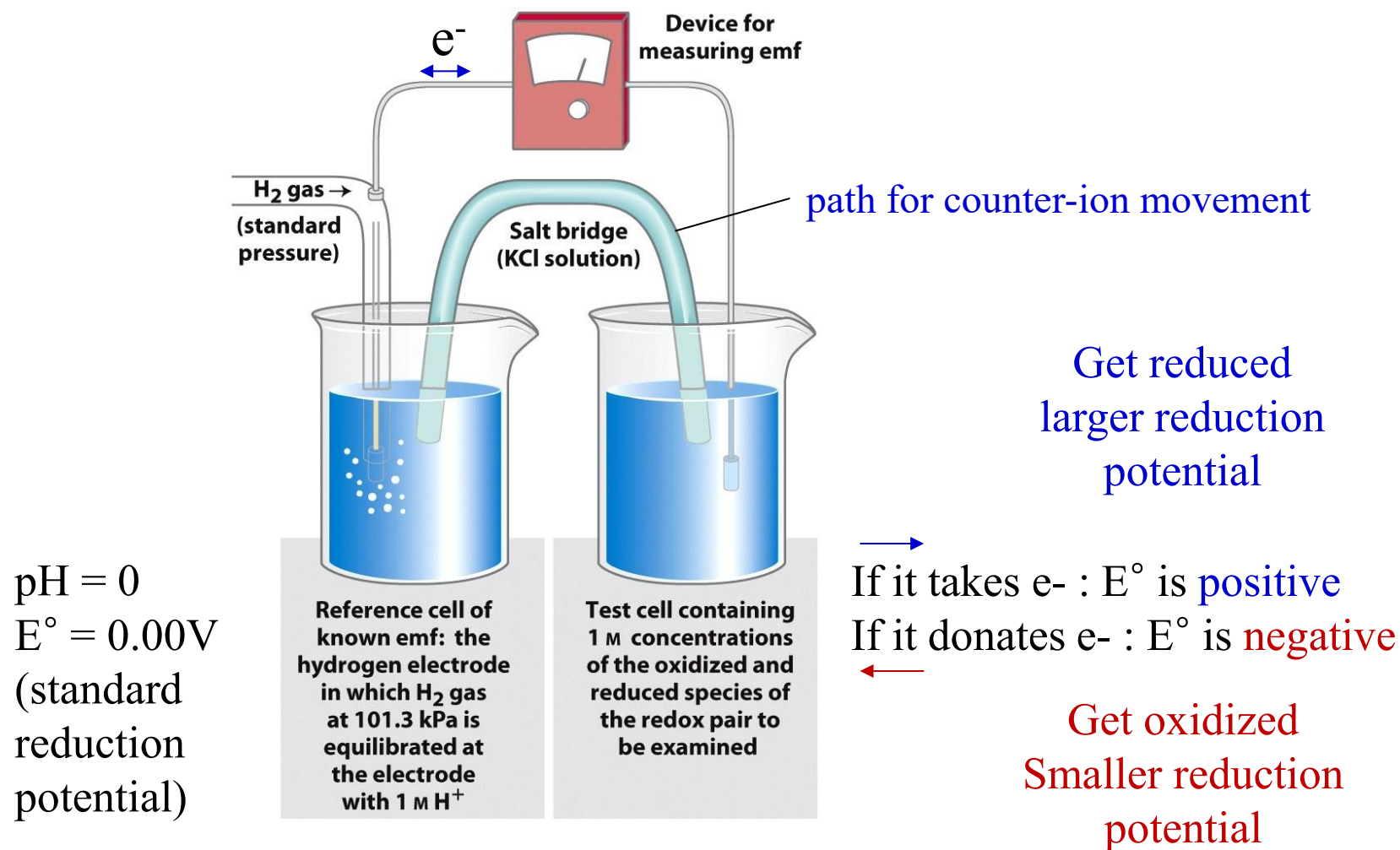
(3) As a hydride ion (:H-)

NAD-linked dehydrogenase

(4) Through direct combination with oxygen



Measurement of the standard reduction potential (E°) of a redox pair



One can find emf
of the test cell

TABLE 13–7

Standard Reduction Potentials of Some Biologically Important Half-Reactions

Half-reaction	E'° (V)	Half-reaction	E'° (V)
$\frac{1}{2}\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{O}$	0.816	$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$ (at standard conditions, pH 0)	0.000
$\text{Fe}^{3+} + \text{e}^- \longrightarrow \text{Fe}^{2+}$	0.771	$\text{Crotonyl-CoA} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{butyryl-CoA}$	−0.015
$\text{NO}_3^- + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{NO}_2^- + \text{H}_2\text{O}$	0.421	$\text{Oxaloacetate}^{2-} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{malate}^{2-}$	−0.166
$\text{Cytochrome } f (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } f (\text{Fe}^{2+})$	0.365	$\text{Pyruvate}^- + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{lactate}^-$	−0.185
$\text{Fe} (\text{CN})_6^{3-} (\text{ferricyanide}) + \text{e}^- \longrightarrow \text{Fe} (\text{CN})_6^{4-}$	0.36	$\text{Acetaldehyde} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{ethanol}$	−0.197
$\text{Cytochrome } a_3 (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } a_3 (\text{Fe}^{2+})$	0.35	$\text{FAD} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{FADH}_2$	−0.219*
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{O}_2$	0.295	$\text{Glutathione} + 2\text{H}^+ + 2\text{e}^- \longrightarrow$ $2 \text{ reduced glutathione}$	−0.23
$\text{Cytochrome } a (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } a (\text{Fe}^{2+})$	0.29	$\text{S} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2\text{S}$	−0.243
$\text{Cytochrome } c (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } c (\text{Fe}^{2+})$	0.254	$\text{Lipoic acid} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{dihydrolipoic acid}$	−0.29
$\text{Cytochrome } c_1 (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } c_1 (\text{Fe}^{2+})$	0.22	$\text{NAD}^+ + \text{H}^+ + 2\text{e}^- \longrightarrow \text{NADH}$	−0.320
$\text{Cytochrome } b (\text{Fe}^{3+}) + \text{e}^- \longrightarrow$ $\text{cytochrome } b (\text{Fe}^{2+})$	0.077	$\text{NADP}^+ + \text{H}^+ + 2\text{e}^- \longrightarrow \text{NADPH}$	−0.324
$\text{Ubiquinone} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{ubiquinol} + \text{H}_2$	0.045	$\text{Acetoacetate} + 2\text{H}^+ + 2\text{e}^- \longrightarrow$ $\beta\text{-hydroxybutyrate}$	−0.346
$\text{Fumarate}^{2-} + 2\text{H}^+ + 2\text{e}^- \longrightarrow \text{succinate}^{2-}$	0.031	$\alpha\text{-Ketoglutarate} + \text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow$ isocitrate	−0.38
		$2\text{H}^+ + 2\text{e}^- \longrightarrow \text{H}_2$ (at pH 7)	−0.414
		$\text{Ferredoxin} (\text{Fe}^{3+}) + \text{e}^- \longrightarrow \text{ferredoxin} (\text{Fe}^{2+})$	−0.432

We can predict the direction of electron flow : **from lower E'° to higher E'°**

Strength of the flow tendency : is proportional to ΔE

$$\Delta G = -n\mathcal{F} \Delta E$$

$$\Delta G'^{\circ} = -n\mathcal{F} \Delta E'^{\circ}$$

The actual reduction potential E is,

$$E = E^{\circ} + \frac{RT}{n\mathcal{F}} \ln \frac{[electron\ acceptor]}{[electron\ donor]}$$

$$E = E^{\circ} + \frac{0.0026\text{ V}}{n} \ln \frac{[electron\ acceptor]}{[electron\ donor]}$$

E° : standard reduction potential

E'° : transformed standard reduction potential

(at pH 7, 25°C)

$\Delta E'^{\circ}$: E'° of the electron acceptor – E'° of the electron donor

$\mathcal{F}$: faraday's constant

TABLE 13-1**Some Physical Constants and Units
Used in Thermodynamics**

Boltzmann constant, $k = 1.381 \times 10^{-23} \text{ J/K}$
Avogadro's number, $N = 6.022 \times 10^{23} \text{ mol}^{-1}$
Faraday constant, $\mathcal{F} = 96,480 \text{ J/V}\cdot\text{mol}$
Gas constant, $R = 8.315 \text{ J/mol}\cdot\text{K}$
($= 1.987 \text{ cal/mol}\cdot\text{K}$)

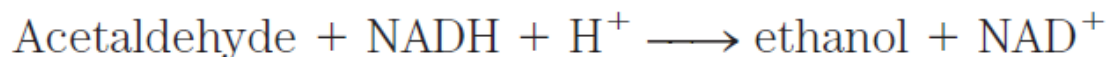
Units of ΔG and ΔH are J/mol (or cal/mol)
Units of ΔS are $\text{J/mol}\cdot\text{K}$ (or $\text{cal/mol}\cdot\text{K}$)
 $1 \text{ cal} = 4.184 \text{ J}$

Units of absolute temperature, T , are Kelvin, K
 $25^\circ\text{C} = 298 \text{ K}$
At 25°C , $RT = 2.478 \text{ kJ/mol}$
($= 0.592 \text{ kcal/mol}$)

WORKED EXAMPLE 13–3

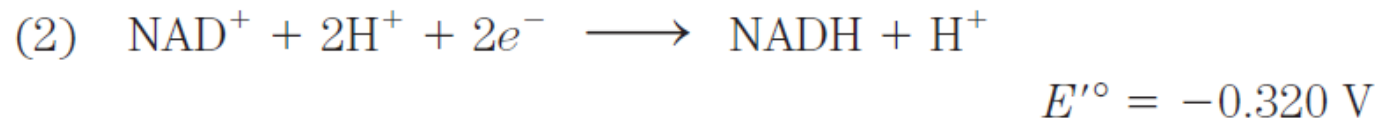
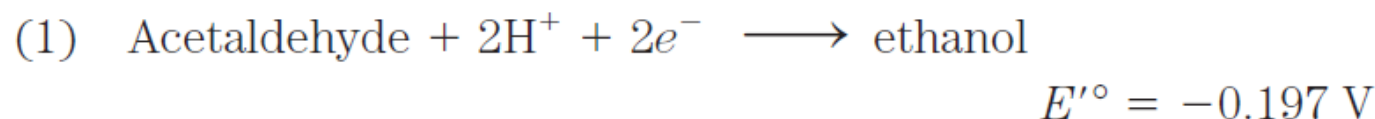
Calculation of $\Delta G'^{\circ}$ and ΔG of a Redox Reaction

Calculate the standard free-energy change, $\Delta G'^{\circ}$, for the reaction in which acetaldehyde is reduced by the biological electron carrier NADH:



Q

Then calculate the *actual* free-energy change, ΔG , when [acetaldehyde] and [NADH] are 1.00 M, and [ethanol] and [NAD⁺] are 0.100 M. The relevant half-reactions and their E'° values are:



Remember that, by convention, $\Delta E'^{\circ}$ is E'° of the electron acceptor minus E'° of the electron donor.

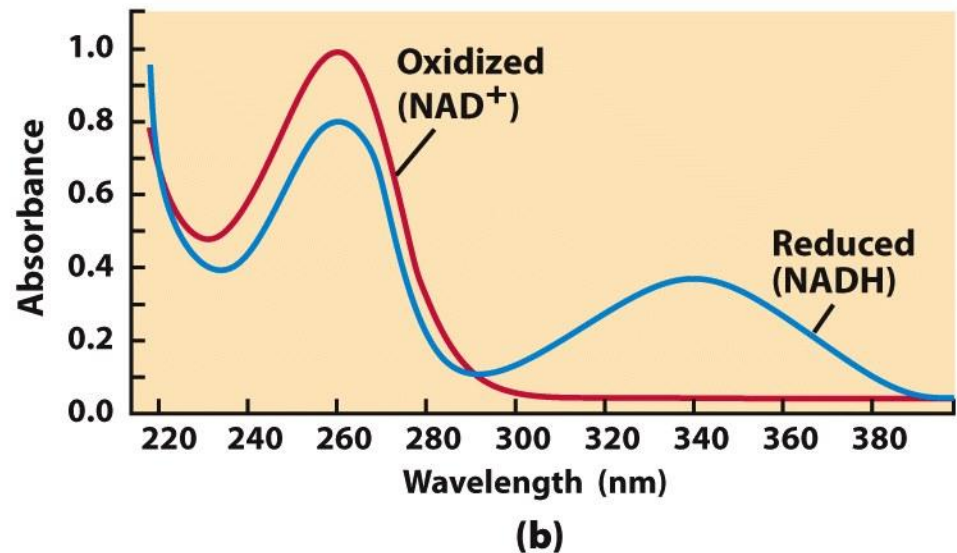
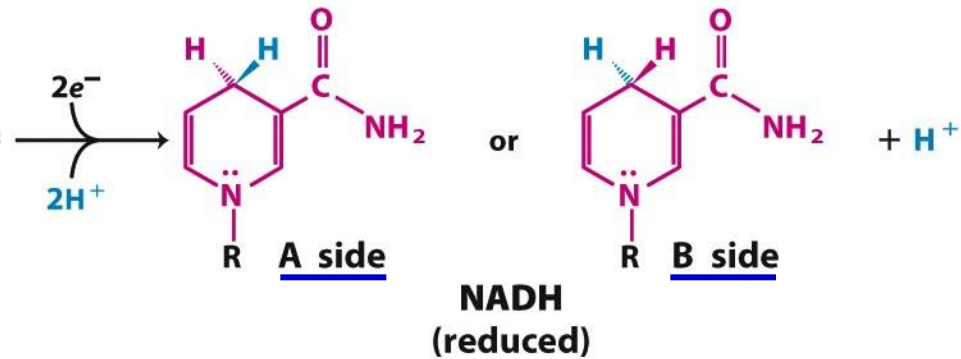
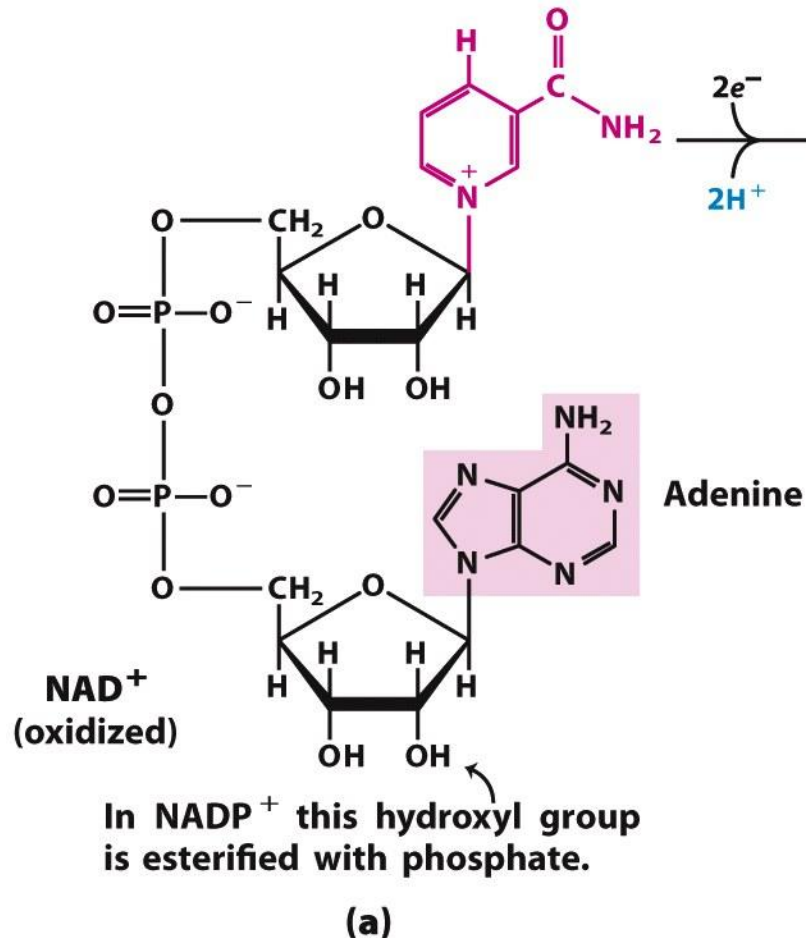
ATP is the energy currency (or carrier) in our body.

What are the electron carriers in our body ?

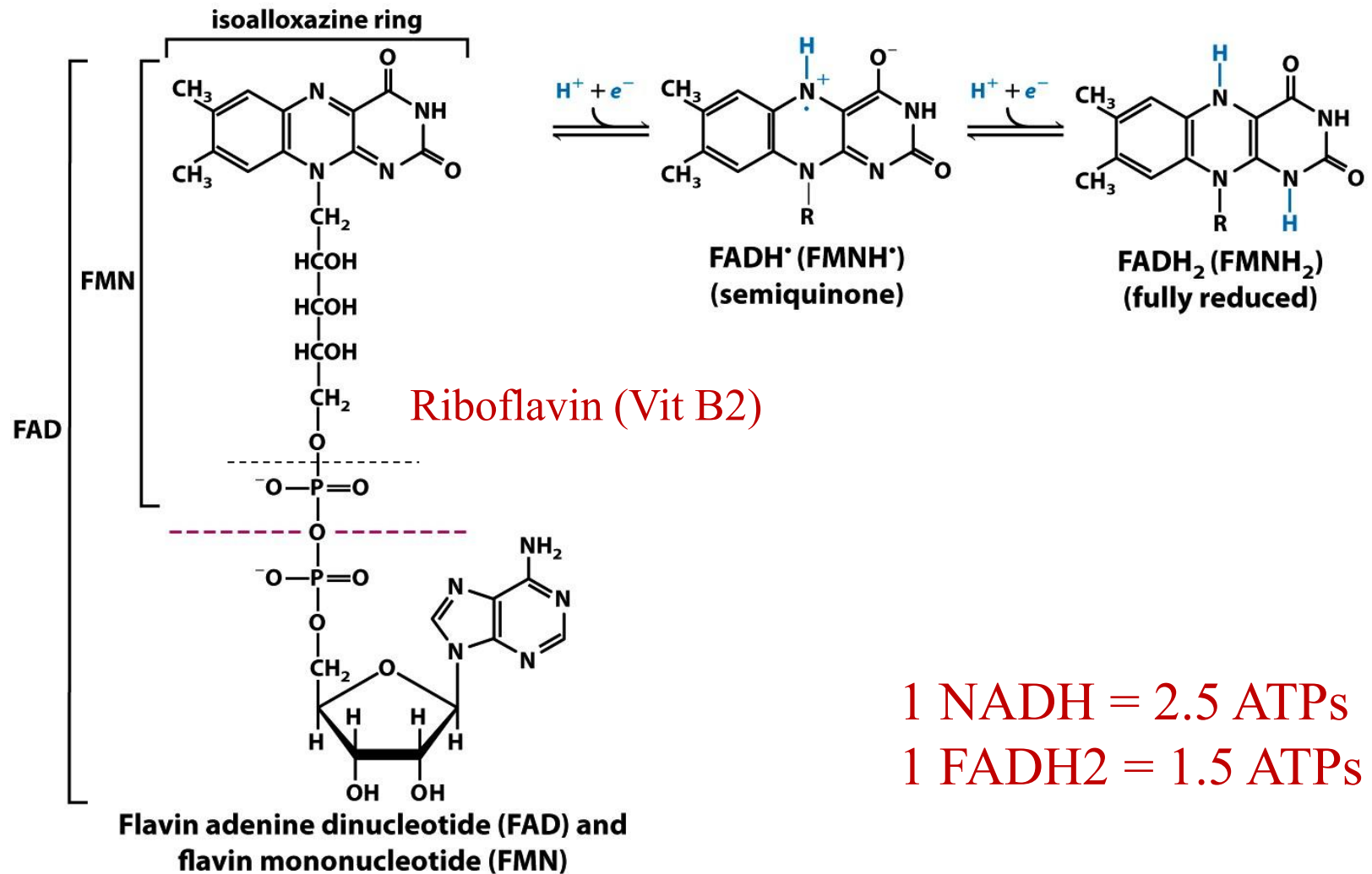
NAD and NADP

+ indicate the oxidized state of nicotinamide ring
 “H” denotes the added hydride ion

The 2nd H⁺ is released



Oxidized and reduced FAD and FMN



1 NADH = 2.5 ATPs
1 FADH₂ = 1.5 ATPs

Figure 13-27

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Role of Coenzymes

Cellular respiration of glucose to CO_2 is carried out in **a series of controlled reactions including oxidations**

Electrons removed from these oxidation steps → to **coenzymes**

Coenzymes : work as electron carrier

NAD, NADP / FMN, FAD / ubiquinone, plastoquinone
iron-sulfur proteins, cytochromes

Notes

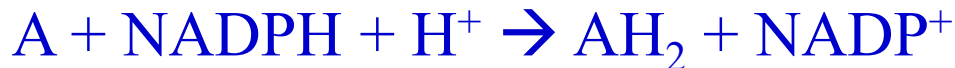
The total conc. of $\text{NAD}^+ + \text{NADH} : 10^{-5} \text{ M}$
 $\text{NADP}^+ + \text{NADPH} : 10^{-6} \text{ M}$

In general, NAD^+ (oxidized) $>$ NADH (reduced)
 NADPH (reduced) $>$ NADP^+ (oxidized)

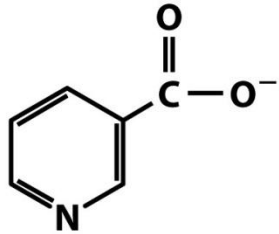
NAD^+ : works in oxidations in catabolic reaction
(in mitochondria)



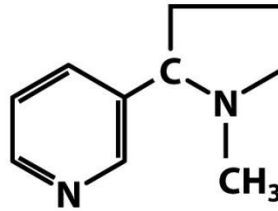
NADPH : works in reduction in anabolic reaction
(in cytosol)



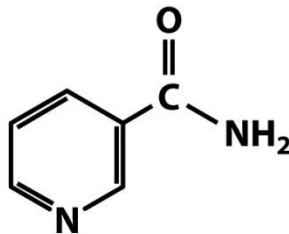
Trp → Niacin → nicotinamide of NAD or NADP



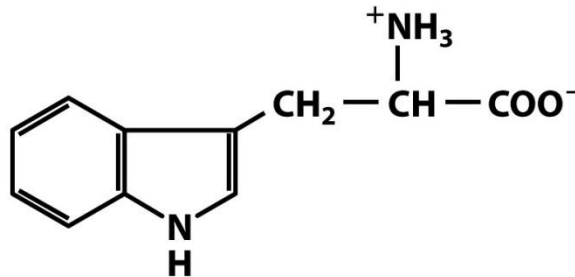
Niacin
(nicotinic acid)



Nicotine



Nicotinamide



Tryptophan

Figure 13-26
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Lack of Trp (corn) → niacin deficiency → pellagra
→ dermatitis, diarrhea and dementia

Summary

1. Cells require sources of free energy (ex. ATP)
2. Standard free-energy change is directly related to the equilibrium constant
3. Actual free-energy changes depend on reactant and product concentrations
4. 5 common biochemical reactions
5. ATP provides energy by group transfers, not by simple hydrolysis
6. Assembly of macromolecules, active transport, and muscle contraction, require ATP energy
7. The flow of electrons can do biological work
8. Oxidation is losing e^- and H , reduction is gaining.
9. Standard reduction potential can be used to calculate free-energy change
10. Cellular oxidation of glucose to CO_2 requires specialized electron carriers (NAD^+ , FAD, FMN, etc)

Factors affecting the actual free-energy change of ATP hydrolysis

1. Concentrations of all reactants and products
2. pH, temperature, $[\text{Mg}^{2+}]$
3. The total concentration of ATP, ADP, P_i and H^+
: higher than free concentrations (due to protein bound form)

**** Cellular ATP concentrations are far above the equilibrium**
- Living cells have developed elaborate mechanism to maintain high concentrations of ATP